

D891

Calculus II

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

Method © 2026 Erik Daniel Lindquist — Course Mastery Coach, armored drill. Items drafted with AI assistance under his direction.

How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A physics teacher models the work to move a probe far from a planet using the integral from 1 to infinity of $1/x^{1.5} dx$. The probe was designed by a university team and weighs 400 kg. What is the value of the integral?

- A. $2/3$
- B. 2
- C. It diverges, because $1/x$ diverges.
- D. It diverges, because the integrand never reaches 0.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2. [TRANSFER — reversed stem / different variable]

Which of these improper integrals from 1 to infinity diverges?

- A. The integral of $1/x^{1.01}$
- B. The integral of $1/\text{square root of } x$
- C. The integral of $1/x^3$
- D. The integral of $1/x^2$

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

An engineer at a water plant estimates total flow with the trapezoidal rule using hourly readings. The plant was built in 1998 and serves 40,000 homes. The flow-rate function is concave up all day. How does the trapezoidal estimate compare with the true total?

- A. It underestimates.
- B. It is exact, because hourly readings are enough.
- C. It overestimates only if the flow is increasing.
- D. It overestimates, because the trapezoid tops lie above a concave-up curve.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

A student in a math club that meets during lunch tests the series sum of $n / (2n + 1)$ for $n = 1$ to infinity. The club has 15 members. What can be concluded?

- A. The nth-term test is inconclusive.
- B. It converges, because the terms are bounded.
- C. It diverges by the nth-term test.
- D. It converges to $1/2$.

Target word _____	The one fact that decides _____	Answer ____	☐ sure ☐ unsure ☐ guess
-----------------------------	---	-----------------------	---

5.

A drug model in a pharmacy class says each dose leaves $2/3$ of the previous amount in the body. The class uses a new textbook. The total amount is the sum from $n = 0$ to infinity of $6(2/3)^n$ mg. What is the sum?

- A. It diverges.
- B. 12 mg
- C. 9 mg
- D. 18 mg

Target word _____	The one fact that decides _____	Answer ____	☐ sure ☐ unsure ☐ guess
-----------------------------	---	-----------------------	---

6. [TRANSFER — reversed stem / different variable]

A geometric series has first term 5 and sum 20. What is the common ratio?

- A. $1/4$
- B. 4
- C. $-3/4$
- D. $3/4$

Target word _____	The one fact that decides _____	Answer ____	☐ sure ☐ unsure ☐ guess
-----------------------------	---	-----------------------	---

7.

A tutor at a community college writes the power series sum of $n x^n / 4^n$. The tutor has taught the course for six years. What is the radius of convergence?

- A. 1
- B. 4
- C. $1/4$
- D. Infinity

Target word _____	The one fact that decides _____	Answer ____	☐ sure ☐ unsure ☐ guess
-----------------------------	---	-----------------------	---

8. [TRANSFER — reversed stem / different variable]

A power series centered at $x = 2$ has radius of convergence 3. Which x value is guaranteed to be in the interval of convergence?

- A. Every real x
- B. $x = -2$
- C. $x = 4$
- D. $x = 6$

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

In a lesson on approximating growth, a teacher asks for the coefficient of x^3 in the Maclaurin series of $e^{(2x)}$. She uses a slide deck from last year. What is it?

- A. $4/3$
- B. $8/3$
- C. 8
- D. $1/6$

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10. [TRANSFER — reversed stem / different variable]

The Maclaurin series of f has coefficient 5 on x^4 . What is $f''''(0)$?

- A. 120
- B. $5/24$
- C. 20
- D. 5

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

A biology club at a rural high school models a quantity with $dy/dx = 2xy$ and $y(0) = 5$. The club meets in the library. What is the solution?

- A. $y = 5e^{(2x)}$
- B. $y = e^{(x^2)} + 5$
- C. $y = 5e^{(x^2)}$
- D. $y = 5 + x^2$

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

A robot arm in a school makerspace moves along $x = t^2$, $y = t^3 - 3t$. The makerspace has a 3D printer and a laser cutter. What is the slope dy/dx at $t = 2$?

- A. $9/4$
- B. 9
- C. 2
- D. $4/9$

Target word	The one fact that decides	Answer	☐ sure ☐ unsure ☐ guess

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer B Concept 1

A physics teacher models the work to move a probe far from a planet using the integral from 1 to infinity of $1/x^{1.5} dx$. ~~The probe was designed by a university team and weighs 400 kg.~~ What is the value of the integral?

Target word: evaluate the improper integral **Decides:** $p = 1.5 > 1$, so it converges to $1/(p - 1)$.

A	Inverted operation	Uses $1/p$ instead of $1/(p - 1)$.
B	Correct	The antiderivative is $-2x^{(-0.5)}$; the limit gives $0 - (-2) = 2$.
C	Confusion set	The p-test depends on p ; here $p > 1$.
D	Absolute language	An integrand only needs to shrink fast enough, not reach 0.

2. Answer B Concept 1 · transfer

Which of these improper integrals from 1 to infinity diverges?

Target word: find the divergent one **Decides:** $p \leq 1$ diverges.

A	Confusion set	p just above 1 still converges.
B	Correct	$p = 1/2 \leq 1$.
C	Inverted operation	$p = 3$ converges to $1/2$.
D	Inverted operation	$p = 2 > 1$ converges to 1.

3. Answer D Concept 2

An engineer at a water plant estimates total flow with the trapezoidal rule using hourly readings. ~~The plant was built in 1998 and serves 40,000 homes.~~ The flow-rate function is concave up all day. How does the trapezoidal estimate compare with the true total?

Target word: direction of error **Decides:** Concave up means chords lie above the curve.

A	Inverted operation	That is the midpoint rule for concave up.
B	Absolute language	Error depends on concavity, not reading frequency alone.
C	Confusion set	Increasing or decreasing decides left/right sums, not trapezoid error.
D	Correct	For $f'' > 0$, trapezoids sit above the graph.

4. Answer C Concept 3

A student in a math club ~~that meets during lunch~~ tests the series sum of $n / (2n + 1)$ for $n = 1$ to infinity. ~~The club has 15 members.~~ What can be concluded?

Target word: convergence **Decides:** Terms approach $1/2$, not 0.

A	Adjacent category	Inconclusive only when the terms go to 0.
B	Dropped element	Bounded terms are not enough; they must approach 0 at least.
C	Correct	The limit of the terms is $1/2$, which is not 0.
D	Confusion set	$1/2$ is the limit of the terms, not the sum.

5. Answer D Concept 4

A drug model in a pharmacy class says each dose leaves $\frac{2}{3}$ of the previous amount in the body. ~~The class uses a new textbook.~~ The total amount is the sum from $n = 0$ to infinity of $6(\frac{2}{3})^n$ mg. What is the sum?

Target word: geometric sum **Decides:** First term 6, $r = \frac{2}{3}$.

A	Absolute language	$ r < 1$, so it converges.
B	Dropped element	Starts at $n = 1$, dropping the first term of 6.
C	Inverted operation	Divides 6 by r instead of by $1 - r$.
D	Correct	$6 / (1 - \frac{2}{3}) = 18$.

6. Answer D Concept 4 • transfer

A geometric series has first term 5 and sum 20. What is the common ratio?

Target word: solve for r **Decides:** $5 / (1 - r) = 20$.

A	Inverted operation	That is $1 - r$, not r .
B	Wrong stem number	The ratio $20/5$; $ r $ must be less than 1 to converge.
C	Confusion set	A negative ratio gives a sum of $5 / 1.75$, about 2.86.
D	Correct	$1 - r = 1/4$, so $r = 3/4$.

7. Answer B Concept 5

A tutor at a community college writes the power series sum of $n x^n / 4^n$. ~~The tutor has taught the course for six years.~~ What is the radius of convergence?

Target word: R **Decides:** Ratio test gives $|x|/4 < 1$.

A	Dropped element	Drops the 4^n in the denominator.
B	Correct	The limit of the ratio is $ x /4$, so $ x < 4$.
C	Inverted operation	Inverts the radius.
D	Absolute language	The terms grow for $ x > 4$.

8. Answer C Concept 5 • transfer

A power series centered at $x = 2$ has radius of convergence 3. Which x value is guaranteed to be in the interval of convergence?

Target word: test a point **Decides:** Converges for $|x - 2| < 3$, that is $-1 < x < 5$.

A	Absolute language	$R = 3$ is finite.
B	Dropped element	Uses $ x < 3$, forgetting the center.
C	Correct	$ 4 - 2 = 2 < 3$.
D	Confusion set	Treats 3 as covering too wide an interval; $ 6 - 2 = 4 > 3$.

9. Answer A Concept 6

In a lesson on approximating growth, a teacher asks for the coefficient of x^3 in the Maclaurin series of $e^{(2x)}$. She uses a slide deck from last year. What is it?

Target word: coefficient **Decides:** $f'''(0) / 3!$ with $f'''(0) = 8$.

A	Correct	$8 / 6 = 4/3$.
B	Wrong stem number	Divides by 3 instead of $3!$.
C	Dropped element	Forgets to divide by $3!$.
D	Dropped element	Forgets the 2^3 from the chain rule.

10. Answer A Concept 6 · transfer

The Maclaurin series of f has coefficient 5 on x^4 . What is $f''''(0)$?

Target word: solve for the derivative **Decides:** $f''''(0) = 5 \times 4!$.

A	Correct	$5 \times 24 = 120$.
B	Inverted operation	Divides by $4!$ instead of multiplying.
C	Wrong stem number	Multiplies by 4 instead of $4!$.
D	Dropped element	Treats the coefficient as the derivative.

11. Answer C Concept 7

A biology club at a rural high school models a quantity with $dy/dx = 2xy$ and $y(0) = 5$. The club meets in the library. What is the solution?

Target word: solve the separable DE **Decides:** $dy/y = 2x \, dx$ gives $\ln|y| = x^2 + C$.

A	Dropped element	Treats x as a constant when integrating $2x$.
B	Inverted operation	Adds the constant after exponentiating instead of multiplying.
C	Correct	Exponentiate: $y = Ce^{(x^2)}$; $y(0) = 5$ gives $C = 5$.
D	Confusion set	Integrates as if $dy/dx = 2x$, dropping the y .

12. Answer A Concept 8

A robot arm in a school makerspace moves along $x = t^2$, $y = t^3 - 3t$. The makerspace has a 3D printer and a laser cutter. What is the slope dy/dx at $t = 2$?

Target word: parametric slope **Decides:** $dy/dt = 9$ and $dx/dt = 4$ at $t = 2$.

A	Correct	$(3t^2 - 3) / (2t)$ at $t = 2 = 9/4$.
B	Dropped element	Uses only dy/dt .
C	Wrong stem number	That is $y(2)$, not a slope.
D	Inverted operation	Divides dx/dt by dy/dt .

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	9	
Wrong stem number	A figure copied from the prompt that is not the answer.	4	
Adjacent category	The right idea in the wrong bucket.	1	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	10	
Absolute language	Always, never, guarantee — where judgment applies.	5	
Confusion set	Every option is real; the test is exact role-matching.	7	
Camouflage	Answers a stem fact that was never the question.	0	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Improper integrals and the p-test	The integral from 1 to infinity of $1/x^p$ converges when $p > 1$ (to $1/(p - 1)$) and diverges when $p \leq 1$.	Thinking any integrand that goes to 0 gives a finite area.	An integral to infinity of a power of x .
2	Trapezoid vs. midpoint error	For a concave-up function the trapezoidal rule overestimates and the midpoint rule underestimates; concave down reverses both.	Assuming both rules err in the same direction, or tying error direction to increasing/decreasing.	The sign of f'' and a question about over- or underestimate.
3	The nth-term (divergence) test	If the terms do not approach 0, the series diverges; if they do approach 0, the test says nothing.	Concluding a series converges because its terms go to 0, or that it converges to the limit of its terms.	A series whose terms approach a nonzero number.
4	Geometric series sum	A geometric series with $ r < 1$ converges to (first term) / $(1 - r)$; it diverges if $ r \geq 1$.	Using the wrong first term when the index starts at 1, or using a/r .	A constant ratio between terms and a question about the sum.
5	Radius and interval of convergence	The ratio test gives a radius R ; a power series centered at c converges for $ x - c < R$, with endpoints checked separately.	Forgetting the center, or inverting the ratio to get R .	A power series with $x - c$ and powers of a constant in the terms.
6	Taylor (Maclaurin) coefficients	The coefficient of $(x - c)^n$ is $f^{(n)}(c) / n!$.	Forgetting the $n!$ or the power from the chain rule.	Asked for a specific coefficient, or a derivative value from a series.
7	Separable differential equations	Separate variables, integrate both sides, and apply the initial condition; the constant multiplies after exponentiating.	Adding the initial value as a constant after exponentiating, or treating x as a constant.	dy/dx equals a product of an x -part and a y -part, with an initial condition.
8	Slope of a parametric curve	For $x(t)$, $y(t)$, $dy/dx = (dy/dt) / (dx/dt)$.	Dividing in the wrong order, or using only dy/dt .	A curve given by x and y in terms of t .

Course scope (public description): Calculus II builds upon foundational calculus concepts to deepen learners' understanding of integration techniques, sequences, series, and differential equations. The course begins with advanced integration methods, including improper integrals and numerical approximation techniques, crucial for solving complex real-world problems in fields such as physics and engineering. Students will explore sequences and series, applying convergence tests and power series expansions to analyze and represent functions accurately. Additionally, the course covers differential equations, parametric equations, and polar coordinates, providing essential tools for modeling dynamic systems and solving higher-level mathematical problems. Throughout the course, practical applications in various scientific and engineering disciplines are emphasized, ensuring students develop strong analytical and problem-solving skills. This course prepares learners for advanced studies in mathematics and related fields and prepares students for the more complex applications found in Multivariable Calculus.